

Quantum Safe?

Everywhere you look, someone is now selling “quantum-safe” security. Vendors promise algorithms that will stand up to tomorrow’s quantum computers just as today’s tools withstand laptops and botnets. But there’s an awkward truth underneath that promise: we don’t actually know if any algorithm is *truly* quantum safe, in the sense of being unbreakable no matter how powerful future quantum machines become.

In practice, “quantum-safe” or “quantum-resistant” really means “we don’t know any fast classical or quantum attacks right now.” It does not guarantee permanent safety. Cryptography has always relied on problems that seem hard: factoring large numbers, solving discrete logs and so on. Quantum computing doesn’t give us proofs; it simply changes which problems are in danger. Shor’s algorithm shows that factoring and discrete logs can be cracked efficiently on a big quantum computer, which makes RSA and elliptic-curve cryptography poor choices.

Symmetric cryptography looks a bit better. Quantum algorithms like Grover’s can speed up brute-force key search, but only by a square-root factor. The simple rule is: a quantum attacker makes a 128-bit key feel more like a 64-bit key. The straightforward response is to use larger keys. Schemes like AES-256 and modern hash functions are generally viewed as “quantum-resilient enough” for realistic threats.

The real turbulence is around public-key cryptography, the part that lets strangers agree on secrets and verify software updates. Here, post-quantum cryptography comes in. Lattice-based schemes such as Kyber for key establishment and Dilithium and Falcon for signatures have been standardized and are starting to appear in real systems. They rely on problems like Learning With Errors and finding short vectors in high-dimensional lattices, for which no efficient classical or quantum attacks are known. Other families, like code-based and hash-based schemes, sit alongside them with different trade-offs in speed, key size and design conservatism. All of these, though, are informed guesses



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rather than certainties. Many cryptographers feel more comfortable with hash-based signatures, because they are built from primitives that have been studied for decades. Code-based schemes like Classic McEliece are long-lived, but come with very large public keys. By contrast, isogeny-based cryptography, seen as an elegant post-quantum path, has been badly broken in public. That episode is a reminder that “no known attack” does not mean “no attack exists,” and that new quantum algorithms could still surprise us.

What about the ideal of information-theoretic security i.e. protection that holds even against an all-powerful quantum attacker? On paper, we do have examples. The one-time pad, if used perfectly with truly random keys as long as the message and never reused, is provably unbreakable in both classical and quantum worlds. The catch is that this is almost impossible to scale: generating, sharing and managing that much randomness is not practical for the internet. Quantum Key Distribution (QKD) promises security based on physics, but it needs special hardware, has distance and deployment limits, and depends on real devices that can leak information.

We do not have proofs that factoring is hard, or that lattice problems resist every possible quantum attack. Cryptography works more like an experimental science: people propose schemes, the community tries to break them, and those that survive long enough are called “safe for now.” Quantum computing makes the game complicated, but that basic pattern is the same.

For policymakers, CISOs and product teams, this is unsettling, but it also sharpens the real goal. The aim is not to discover a magical algorithm that is “truly quantum safe” forever. It is to push risk down to a level you can live with. That means phasing out RSA and ECC, adopting standardized post-quantum algorithms and, importantly, building systems that are crypto-agile i.e. able to switch algorithms without a total rebuild. “Quantum-resilient today, and ready to adapt tomorrow” may be the only kind of safety we can realistically achieve. ■